Sprint 3 — Simulation Engine Plan (v3)

Deloitte Capstone: AI for Equitable Public Transportation, Miami-Dade County
April 9, 2026 | Supersedes Sprint3_Plan_v2.pdf (Daniel, Apr 3)

This document is the canonical Sprint 3 reference. It incorporates all design decisions made during Sprint 3.1 (Apr 9) and replaces the v2 plan. Daniel's v2 PDF is preserved in the repo for version history.

Overview

Goal: Build a simulation engine that lets transit planners adjust service parameters and see how equity outcomes change across 504 Miami-Dade census tracts.

Deadline: April 17, 2026

Primary Deloitte deliverable: Sprint 3.3 Scenario Validation Notebook

Sub-Sprint	Dates	Status	Key Output
3.1 Baseline State + Lever Catalog	Apr 9	DONE	Sprint3_Baseline_State.csv Sprint3_Lever_Catalog.json
3.2 simulator.py Module	Apr 10-13	NEXT	simulator.py + test_simulator.py
3.3 Scenario Validation Notebook	Apr 14-16	PENDING	Sprint3_Scenario_Validation.ipynb Sprint3_Scenario_Results.csv
3.4 Interactive Prototype (Gradio)	Post Apr 17	STRETCH	Sprint_3_Gradio_V3.ipynb

Architecture

Single-model pipeline. The user adjusts transit-service levers; the need side is held static.

```
User adjusts 4 levers --> modify 13-feature matrix --> XGBoost v3 --> new deficit_predicted
```

```
ratio = new_deficit / baseline_deficit
```

```
new_equity = composite_need x (composite_access_deficit x ratio)
```

```
new_tier = Sprint 2a fixed cutoffs
```

Why proportional-change (not naive multiplication)?

- The v3 model predicts **service_deficit** (5-indicator index) -- a different construct from Sprint 2a's **composite_access_deficit** (7-indicator composite). Correlation is only 0.61.
- The ACS time-series produces **projected_need_2027** (0-100 scale) -- a different construct from Sprint 2a's **composite_need** (0-1 scale). Rank correlation is only 0.67.
- Naive multiplication of these mismatched measures makes Sprint 2a tier cutoffs meaningless. Testing confirmed: **only 57% tier recovery at zero delta**.
- Proportional-change approach: XGBoost v3 computes a **ratio of deficit change**, applied to the original Sprint 2a composite_access_deficit. At zero delta, ratio = 1 and equity = baseline exactly. **99.6% tier recovery**.

- **composite_need** (Sprint 2a, current, 0-1 scale) is the need side of the equation. **projected_need_2027** stays in the baseline for filtering and context (fragile flags, worsening flags) but does NOT enter the equity multiplication.

Key Design Decisions (Apr 9)

1. Four transit-service levers only

Daniel's original catalog had 8 levers. We trimmed to 4:

Lever	SHAP	Unit	Delta Range	Mean
freq_peak_am_tph	#1 (39.6%)	trips/hr	[-10, +10]	26.7
weekend_weekday_ratio	#2 (21.7%)	ratio	[-0.5, +0.5]	0.43
freq_early_tph	#3 (6.9%)	trips/hr	[-5, +5]	2.98
rail_trip_share	Modal lever	fraction	[-0.1, +0.3]	0.011

What was removed and why:

- **Headway features** (headway_peak_am_min, headway_early_min): physically inverse of frequency (headway = 60/freq). Exposing both allows impossible states. Simulator auto-derives headway internally from the frequency lever.
- **Demographic features** (poverty_rate_pct, hh_no_vehicle_pct): forecast outputs from ACS time-series, not parameters a transit agency controls. A simulator lever must be something a planner can change by signing a work order.

rail_trip_share is not in the SHAP top 6, but what-if analysis shows it is the single most impactful lever: +10pp produces -0.0149 deficit reduction with 98% of tracts improving (roughly 2x the next best lever).

2. OLS need model deferred to v2

- Daniel built an OLS model (R^2=0.817) regressing composite_need on poverty_rate_pct and hh_no_vehicle_pct. Intended for need-side stress tests.
- **Circularity concern:** composite_need was constructed from these same features in Sprint 2a. The OLS partially fits a formula to its own components.
- Not needed in v1: simulator only exposes transit-service parameters. Need side is static from ACS forecast.
- Demographic shock scenarios (e.g., 'what if poverty rises 5pp?') are a legitimate v2 feature. If re-enabled, they should be labeled **'stress tests'** (not levers) and use Sprint 2a formula weights directly instead of OLS.
- Daniel's file preserved at **Sprint 3/future/Sprint3_Need_OLS_Coefficients.json** with README.

3. Zero-frequency tract handling

- 176 tracts (35%) have freq_peak_am_tph = 0 (no peak AM service). 209 tracts (41%) have freq_early_tph = 0.
- In training data, all zero-frequency tracts have headway = 120 min (GTFS pipeline's 'no service' cap).
- Simulator derives headway = 60 / freq. At freq = 0, this is undefined. **Fix:** cap headway at 120 min when post-delta freq is at or below 0. Clamp frequency to 0 (no negatives).

- Adding service to a zero-frequency tract (e.g., +2 tph) produces a large feature swing (headway: 120 to 30 min). This is correct -- going from no service to some service IS a big change. Simulator flags these tracts in output.

4. SHAP ranking correction

- Prior documentation listed headway_peak_am_min as SHAP #3. Actual ranking from the v3 notebook (Apr 6 re-run): **freq_early_tph is #3 (6.9%)**; headway_peak_am_min is #6 (3.8%).
- Full corrected top 6: freq_peak_am_tph (39.6%), weekend_weekday_ratio (21.7%), freq_early_tph (6.9%), trend_commute_wfh_pct (6.2%), trend_commute_public_transit_pct (4.4%), headway_peak_am_min (3.8%).
- Sprint 3.3 scenario validation must use this corrected order when checking that lever impacts are proportional to SHAP importance.

3.1 — Baseline State + Lever Catalog [DONE]

Completed April 9, 2026. Full details in Sprint3_1_Summary.pdf.

Outputs:

File	Description
Sprint3_Baseline_State.csv	504 tracts x 29 cols. 3-way inner join: 13 v3 features + deficit predictions + need projections + equity
Sprint3_Lever_Catalog.json	4 transit-service levers with SHAP ranks, delta ranges, data ranges, derived-feature notes, future-fea
Sprint3_1_Summary.pdf	Teammate-facing document covering all 3.1 decisions and rationale.
Sprint 3/future/	Quarantined OLS coefficients + README for v2 stress-test feature path.

Verification performed:

- All 29 columns trace exactly to source files (zero diff across 504 tracts)
- Equity formula confirmed: composite_need x composite_access_deficit = equity_priority_score (max diff ~1e-17)
- Tier cutoffs monotonically increasing (Low < Moderate < High < Critical, no overlap)
- Zero nulls in all model columns
- All 4 lever data_ranges in catalog match actual data ranges in baseline

> ***Data note:*** *Tract 12086009050 was duplicated 4x (identical rows) in Sprint2b_Tract_Risk_Projections.csv. Deduped before join. Sprint2b_Combined_Risk_Scoring.csv (referenced in v2 plan) was never produced; the baseline state absorbs that join.*

3.2 — Simulator Core Module [Apr 10-13]

Goal: Clean, importable Python module (**simulator.py**) that Sprint 4 dashboard can call directly. No notebook dependencies.

API

```
class Simulator:
```

```
def __init__(self, baseline_path, model_path, scaler_path, lever_catalog_path)
```

```
def run(self, access_deltas={}, tract_filter='all', label='scenario') ->
SimulationResult
```

SimulationResult contains:

- **.tract_df** -- per-tract DataFrame: before/after deficit, equity score, tier
- **.tier_shifts** -- 4x4 matrix (from_tier x to_tier counts)
- **.summary** -- dict: n_tracts_improved, n_tier_upgrades, avg_equity_delta, avg_deficit_delta
- **.label** -- scenario name string

Internal pipeline (4 steps)

Step	Function	What it does
1	_apply_access_delta()	Copy 13-feature matrix. Apply lever deltas to target tracts. Auto-derive headway = 60/freq (cap 12
2	_predict_deficit()	scaler.transform() then xgb.predict(). Clamp output to [0, 1].
3	_compute_equity()	ratio = new_deficit / baseline_deficit. new_equity = composite_need x (composite_access_deficit

4	_diff_report()	Per-tract deltas. 4x4 tier shift matrix. Summary stats. Flag zero-freq tracts that gained service.

Validation gates (test_simulator.py)

- **Zero-delta test:** run() with no deltas. Per-tract equity must match baseline within 1e-5.
- **Isolation test:** apply delta to subset of tracts. Unaffected tracts must be unchanged exactly.
- **Direction tests:** increase freq_peak_am_tph --> avg deficit must fall. Increase rail_trip_share --> avg deficit must fall. Increase weekend_weekday_ratio --> avg deficit must fall.

Outputs: Sprint 3/simulator.py, Sprint 3/test_simulator.py

3.3 — Scenario Validation Notebook [Apr 14-16]

Goal: Run 5 reference scenarios through the simulator. Verify direction, magnitude, and tier logic. This is the **primary Deloitte deliverable** for Sprint 3.

Scenarios (all access-side; need is static in v1)

#	Name	Lever(s)	Scope	Expected Outcome
S1	Peak frequency boost	freq_peak_am_tph +2 trips/hr	Critical tracts (~51)	Deficit drops in Critical; some tier upgrades to High. SHAP #1.
S2	Weekend parity	weekend_weekday_ratio --> 0.80	All 504 tracts	Broad improvement; tracts with low current ratio improve most. SHAP #2.
S3	Early service expansion	freq_early_tph +1 trip/hr	High + Critical (~153)	Shift-worker tracts benefit. SHAP #3. S1 > S3 magnitude.
S4	Rail modal shift	rail_trip_share +0.10	All 504 tracts	Largest single-lever county-wide deficit drop per 2b.5 what-if. 98% of tracts improve.
S5	Combined multi-lever	freq_peak_am +2, weekend --> 0.80, rail +0.05	Critical + High (~153)	Multi-lever stacking; combined delta > any single lever.

Per-scenario reporting

- Per-tract before/after table (deficit, equity score, tier)
- 4x4 tier shift matrix
- Summary: n tracts improved, n tier upgrades/downgrades, avg equity delta, avg deficit delta
- Magnitude sanity check: changes proportional to SHAP importances
- Flag zero-frequency tracts that gained service (outsized improvement expected)

Pass criteria

- All validation gates in test_simulator.py pass
- S1-S3 directions consistent with SHAP rankings (S1 > S3 per-tract deficit delta)
- S4 produces the largest single-lever county-wide deficit improvement (per 2b.5 what-if results)
- S5 combined delta > any single component
- No scenario produces equity scores outside [0, 1] or tiers outside the 4 defined categories

Outputs: Sprint3_Scenario_Validation.ipynb, Sprint3_Scenario_Results.csv

3.4 — Interactive Prototype / Gradio [Stretch]

Priority: V2 prototype already exists (Sprint_3_Gradio_V2.ipynb). If time is tight, 3.3 is the critical deliverable. 3.4 can carry past April 17.

Current state (V2): Choropleth map of 504 tracts, tier filter, tract narrative generator, summary stats. What-if panel operates on a single synthetic tract with median-filled features -- does NOT use simulator.py or the 504-tract baseline.

Upgrades for V3

- **Wire to simulator.py:** replace standalone median-filling logic with Simulator.run() calls. Sliders for 4 levers (freq_peak_am, freq_early, weekend_ratio, rail_share). Tract scope selector (All, Critical, High+Critical, Fragile, custom GEOIDs).
- **Before/after maps:** side-by-side choropleth showing tier changes. Color-code tracts that improved, worsened, or stayed same.
- **Tier shift summary panel:** 4x4 matrix visualization + headline stats (n tracts improved, n tier upgrades).
- **Fragile-tract overlay:** highlight the 22 fragile tracts on the map with a separate marker.

Output: Sprint_3_Gradio_V3.ipynb

Parallel Track: City2Graph

Non-blocking exploration track. Does not touch the simulator pipeline.

- Convert Miami-Dade GTFS into a NetworkX graph via DuckDB (city2graph library). Nodes = stops, edges = service connections with travel times and frequencies.
- Compute network metrics per tract: stop centrality, connectivity, structural isolation.
- Correlate with XGBoost deficit predictions. Evaluate whether graph-derived features could replace or augment spatial features in a future model iteration.
- **Output:** City2Graph_Analysis.ipynb. **Risk:** Low -- standard GTFS format, drop and move on if incompatible.

File Reference

Sprint 3 outputs

File	Description
Sprint3_Baseline_State.csv	504 tracts x 29 cols. Single baseline for simulator. Inner join of features + deficit + need.
Sprint3_Lever_Catalog.json	4 transit-service levers with SHAP ranks, ranges, rationale.
Sprint3_Plan_v3.pdf	This document. Supersedes Sprint3_Plan_v2.pdf.
Sprint3_1_Summary.pdf	Teammate-facing summary of 3.1 decisions.
Sprint 3/future/	Quarantined OLS coefficients + README for v2 stress-test path.
simulator.py (3.2)	Simulator module. Loads baseline, accepts lever deltas, outputs SimulationResult.
test_simulator.py (3.2)	Pytest validation gates: zero-delta, isolation, direction tests.
Sprint3_Scenario_Validation.ipynb (3.3)	Primary Deloitte deliverable. 5 reference scenarios with validation.
Sprint3_Scenario_Results.csv (3.3)	504 tracts x 5 scenario columns.

Sprint 2 inputs

File	Used by
Sprint2b_XGBoost_v3.pkl	simulator.py -- trained XGBoost model for deficit prediction
Sprint2b_Scaler_v3.pkl	simulator.py -- StandardScaler for 13 features
Sprint2b_Modeling_Features_NotebookOutput.csv	3.1 baseline join (13 features + equity context)
Sprint2b_Risk_Scores_v3.csv	3.1 baseline join (deficit_predicted, risk_tier)
Sprint2b_Tract_Risk_Projections.csv	3.1 baseline join (projected_need_2027, fragile flags)
Sprint2b_V3_LeakageFree_Modeling.ipynb	Canonical reference for model, SHAP, what-if results

Version History

Version	Date	Author	Changes
v1	Apr 2	Daniel	Initial plan draft
v2	Apr 3	Daniel	Expanded architecture, day-by-day schedule, Gradio V2 prototype
v3	Apr 9	Luna + Claude	Trimmed to 4 levers (removed headway duplicates + demographic levers). Quarantined OLS to f